Carrousel Filling Systems

Kosan Crisplant's carrousel filling system is designed for safe and effective filling of all kinds of LPG cylinders.

- The most effective way of filling LPG cylinders
- High capacity filling, up to 1,800 cylinders per hour
- Fully automatic cylinder flow
- High safety level thanks to intrinsically safe network
- Decentralized control units independent of PC's
- Rapid encoding or automatic reading of cylinder tare
- Control system uptime: 99% – the best on the market
- Total overview of the filling production thanks to PC data collection

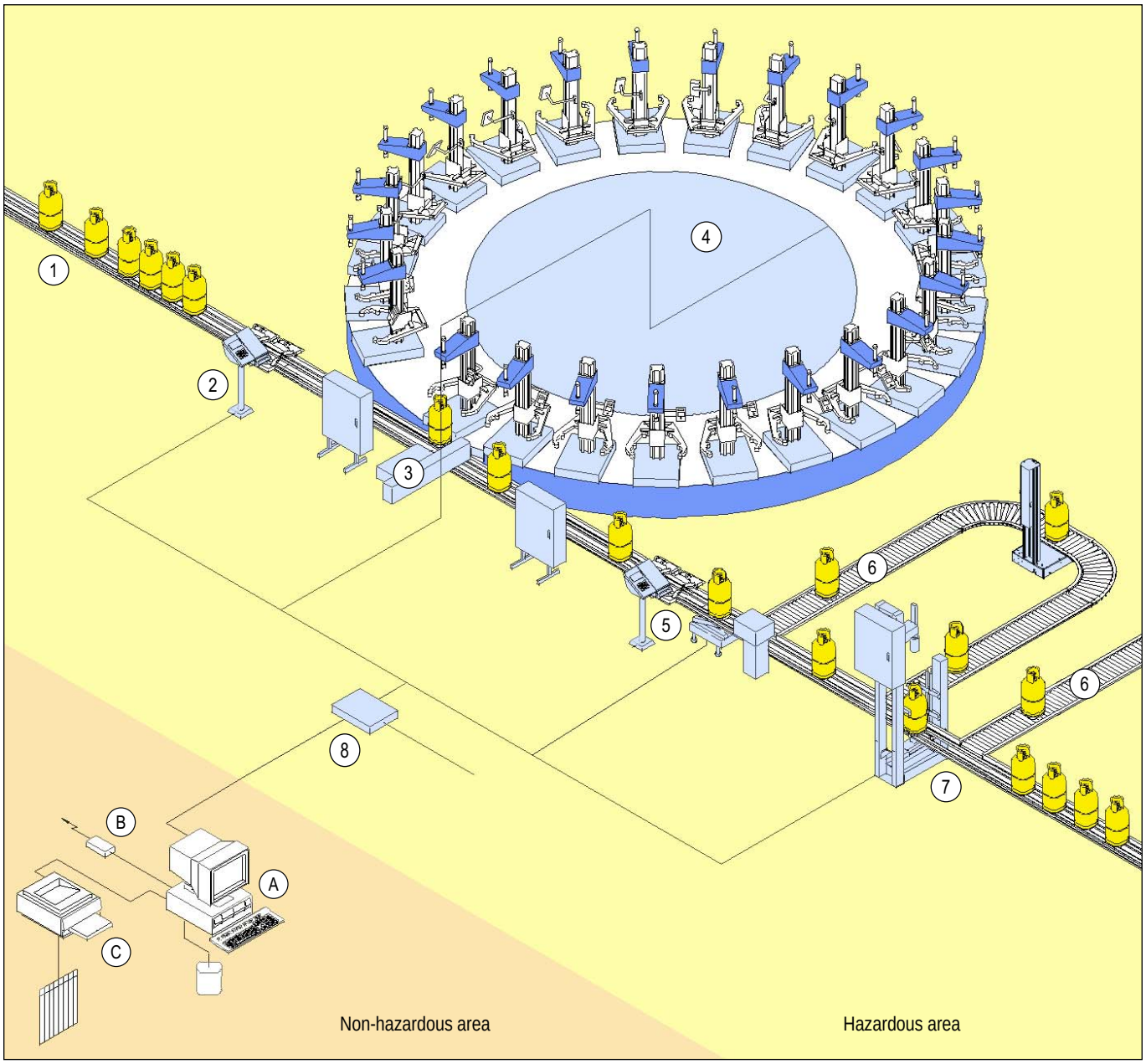
Equipment in hazardous area:

1 Chain conveyor
2 Encoding station
3 Inlet and outlet units
4 Filling carrousel and filling machines
5 Check scale
6 Sort-out conveyor
7 Leak detector
8 Power and data interface

Equipment in non-hazardous area:

A PC incl. software and database for filling data
B Modem for connection to KC on-line service
C Printer for printing reports

Example of carrousel filling system



Carrousel Filling Systems



Carrousel filling system with UFM universal filling machines



Inlet and outlet units



Electronic check scale

Your benefits

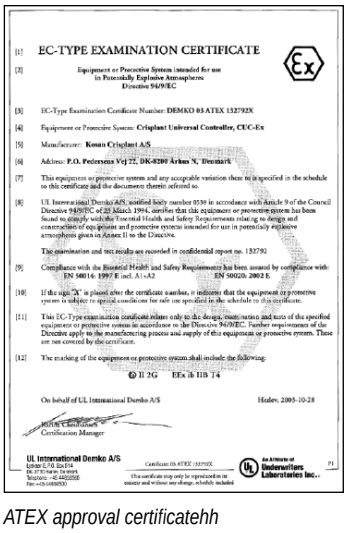
- Low installation costs and high safety thanks to intrinsically safe network
- Focus on ergonomics at central tare encoding station
- Few operators and minimized risk of human errors
- No mechanical wear on electrical control equipment, non-touch sensors and swivel connectors
- Optimal logistics and high output (e.g. sorting of cylinders for maintenance before filling)
- Optimal PC data collection tool for effective filling and maintenance
- Capacity increase is possible
- Same user interface (HMI/CUC) on all machines

Your possibilities

- Semi-automatic or fully automatic system according to valve type
- Automation level can be upgraded according to actual and future needs
- From 8 to 42 filling machines on the carrousel platform – capacity can be adapted to actual and future needs
- Filling machines and filling heads adapted to cylinders and cylinder valves
- PC placed in non-hazardous area can be connected to the system
- Quick and professional service backup by remote control available

Your safety

- All equipment and machines in the filling system are EU approved and designed in accordance with current EU directives EN 50014, EN 50020, EN 50081, EN 50082, EN 55022, incl. the ATEX Directive (94/9/EC)
- All equipment and machines in the filling system are designed for use in hazardous areas classified as Zone 1 according to IEC 79-10 and Class I, Division 1 according to NEC (USA), article 500
- Filling machines and check scales have weighing Accuracy Classification C3 according to OIML R 76/EN45501
- National/local approvals



EC-TYPE EXAMINATION CERTIFICATE

Equipment or Protective System intended for use in Potentially Explosive Atmospheres Directive 94/9/EC

[3] EC-Type Examination Certificate Number: DEMKO 03 ATEX 132792X

[4] Equipment or Protective System: Crisplant Universal Controller, CUC-Ex

[5] Manufacturer: Kosan Crisplant A/S

[6] Address: P.O. Pedersens Vej 22, DK-8200 Århus N, Denmark

[7] This equipment or protective system and any acceptable variation thereto is specified in the schedule to this certificate and the documents therein referred to.

[8] UL International Demko A/S, notified body number 0539 in accordance with Article 9 of the Council Directive 94/9/EC of 23 March 1994, certifies that this equipment or protective system has been found to comply with the Essential Health and Safety Requirements relating to design and construction of equipment and protective systems intended for use in potentially explosive atmospheres given in Annex II to the Directive.

The examination and test results are recorded in confidential report no. 132792

[9] Compliance with the Essential Health and Safety Requirements has been assured by compliance with: EN 50014: 1997 E incl. A1+A2 EN 50020: 2002 E

[10] If the sign 'X' is placed after the certificate number, it indicates that the equipment or protective system is subject to special conditions for safe use specified in the schedule to this certificate.

[11] This EC-Type examination certificate relates only to the design, construction and tests of the specified equipment or protective system in accordance to the Directive 94/9/EC. Further requirements of the Directive apply to the manufacturing process and supply of this equipment or protective system. These are not covered by the certificate.

[12] The marking of the equipment or protective system shall include the following:

⟨Ex⟩ II 2G EEx ib IIB T4

On behalf of UL International Demko A/S — Herlev, 2005-10-28

Certification Manager

UL International Demko A/S

Certificate 03 ATEX 132792X

An Affiliate of Underwriters Laboratories Inc.

ATEX approval certificatehh